

SULTAN MUHAMMAD KHAN

AI / Machine Learning Engineer

Haripur, Pakistan · +92 332 1508021 · akasultan10@gmail.com

linkedin.com/in/sultanmuhammadkhan · github.com/aka-shiro10 · aka-shiro10.github.io/Sultan-portfolio

PROFESSIONAL SUMMARY

Artificial Intelligence student with hands-on experience building end-to-end ML systems from custom-trained CNN and YOLO models to agentic AI systems with autonomous tool use, RAG, voice I/O, and computer vision. Comfortable across Python, FastAPI, Unity (C#), and modern LLM stacks (Groq, Whisper, Kokoro). Strong focus on shipping working prototypes, not just notebooks. Open to ML Engineer, AI Engineer, Agentic AI, and Backend roles remote or on-site.

EDUCATION

BS Artificial Intelligence · Pak-Austria Fachhochschule Institute of Applied Sciences and Technology (PAF-IAST)
Oct 2022 - May 2026

Haripur, Pakistan · Reg. B22F0667AI036 ·

Intermediate (ICS Pre-Engineering) · Jinnah Jame School & College
Haripur, Pakistan
2020 - 2022

Matriculation (Science) · Jinnah Jame School & College
Haripur, Pakistan
2018 - 2020

EXPERIENCE

Data / ML Intern · Sybrid
Islamabad / Remote · 320 hours
May 2025 - Jul 2025

- Built and curated training datasets and ETL pipelines feeding internal ML workflows.
- Set up a vector database and prototyped an internal RAG chatbot using sentence-transformer embeddings and an LLM backend.
- Trained and iterated on the RAG pipeline using a custom bash automation script, enabling repeatable end-to-end experiments without manual setup.

Software Intern · KeyMech
Wah Cantt, Pakistan · 320 hours
Jul 2024 - Sep 2024

- Shipped cross-platform mobile features and REST API integrations used by field maintenance teams.
- Owned production build pipelines and worked with backend engineers to debug live data sync issues.

Audit Intern · Seasons Foods
Lahore, Pakistan · 240 hours
Aug 2023 - Sep 2023

- Performed inventory audits and reconciliations; flagged process gaps that reduced count variances.

FINAL YEAR PROJECT

ENE — 3D AI Desktop Companion
Unity (C#) · Python · FastAPI · WebSocket · Groq LLM · Whisper · Kokoro TTS · YOLOv8 · ONNX · SQLite
Sep 2024 - Present

- Architected a 3D animated anime-style desktop companion that runs as a transparent always-on-top Windows overlay; Unity (C#) client and Python FastAPI backend communicate over LAN via HTTP REST and WebSocket.
- Trained a **custom 7-class emotion CNN** on FER-2013 (66.3% accuracy) deployed via Unity Sentis ONNX with BlazeFace cropping for real-time webcam emotion classification.
- Trained a **YOLOv8s 7-class UI-element detector** reaching 95.85% mAP@0.5 to identify dialogs, taskbars, menus, and active windows on the user's screen in real time.
- Integrated OpenAI Whisper (base.en) STT with noise reduction and HPF, plus Kokoro TTS at 24 kHz with viseme-driven VRM lip-sync - full voice loop with end-to-end latency around 1–2 seconds.
- Built a **15-intent classifier** combining a rule-based pre-check, sentence-transformer cosine similarity, and Groq LLaMA-3.1-8B-instant for free-form replies and fallback resolution.

- Designed a Fernet-encrypted SQLite memory layer (WAL mode) with auto-extracted user facts (regex over speech) and a background daemon that scores personality (formality, humor, technical, verbosity) every 10 turns and adapts ENE's tone.
- Implemented OS-level actions - open / close / minimize apps, switch tabs, web search, screen description — via PowerShell + Win32 P/Invoke from the Unity client, ensuring actions execute on the user's PC, not the server.

SELECTED PROJECTS

Car Racing AI Trainer - *Unity ML-Agents · Reinforcement Learning*

- Trained RL agents on procedurally generated tracks using Unity ML-Agents; experimented with reward shaping for stable lap-time learning.

Evolution Simulator - *Unity · ANN · Genetic Algorithms*

- Built a population-based simulation where agents with feed-forward neural networks evolve foraging behavior under selection pressure.

Network Anomaly Detection - *Python · scikit-learn · Tkinter*

- Trained classical ML models on labeled network traffic to flag anomalous flows; packaged with a simple GUI for live inspection.

Server Parallel Log Analyzer - *Python · Multiprocessing*

- Concurrent log-processing tool using Python multiprocessing to chunk and aggregate large server logs with real-time statistics.

TECHNICAL SKILLS

AI / ML	Python, PyTorch, TensorFlow, ONNX, Hugging Face, scikit-learn, sentence-transformers
LLM Stack	Groq API, LLaMA-3.1 / LLaMA-4-Scout, RAG, vector databases, prompt engineering, intent classification
Agentic AI	Tool / function calling, ReAct, multi-step reasoning, autonomous, MCP,
Computer Vision	YOLOv8, EasyOCR, OpenCV, BlazeFace, Unity Sentis (ONNX inference)
Speech / Audio	OpenAI Whisper, Kokoro TTS, noise reduction, viseme-based lip sync
Backend	FastAPI, WebSocket, REST APIs, SQLite, multiprocessing, Fernet encryption
Languages	Python, C#, C / C++
Frontend / Game	Unity 2022.3 (C#), ML-Agents
Tools / Infra	Git, GitHub, Docker, Linux (Ubuntu), Windows, Azure
Soft Skills	Cross-functional teamwork, technical writing, fast self-learning, ownership of deliverables
Creative / Design	Photoshop, Illustrator, and Premiere Pro

LANGUAGES

Urdu - Native · English - Professional